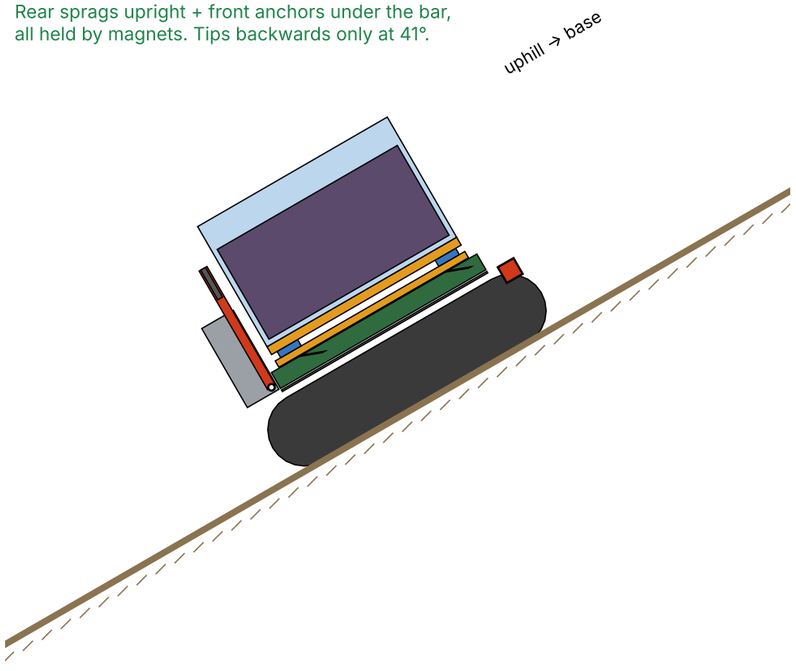
BINBOT — ANTI-SLIDE & ANTI-SPIN PROTECTION ON 30° SLOPES (base uphill: full bins climb)

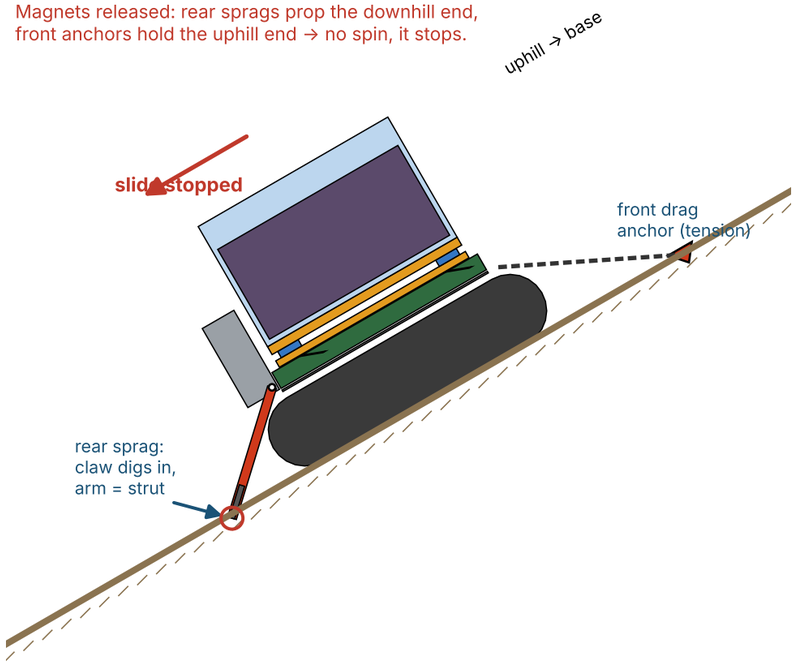
A CLIMBING 30° — NORMAL (400 kg, deck low)

Rear sprags upright + front anchors under the bar,
all held by magnets. Tips backwards only at 41°.

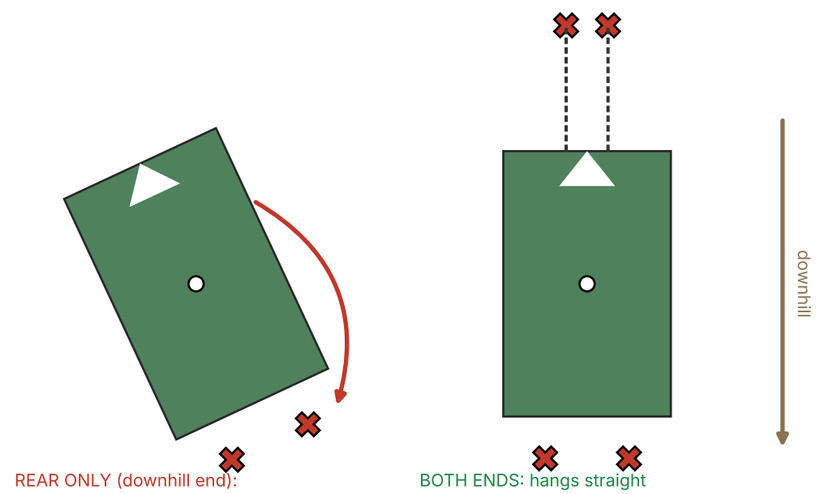


B SLIDE DETECTED — HELD AT BOTH ENDS

Magnets released: rear sprags prop the downhill end,
front anchors hold the uphill end → no spin, it stops.



C WHY IT SPINS — TOP VIEW



REAR ONLY (downhill end):
the heavy body above swings
round the claws → sideways
→ on 30° it ROLLS OVER

BOTH ENDS: hangs straight
below the front anchors like a
pendulum; 4 points 2.5 m apart
stop any rotation

1 PREVENT (before every 30° section)

- Bin ≤ 400 kg, deck LOWERED
- Dry-day only: wet-day mode (rain sensor + farmer switch + weather forecast) blocks 30° routes
- Grip check on the approach slope: slip < 10 %
- Sprags armed: magnet current + position switch OK
- Nobody downhill: pendant GPS positions outside a 20 m corridor below the robot
- Speed 0.3 m/s, straight on the fall line (±5°), no turning
- Going DOWN: in reverse, so the rear (sprags) is downhill (rear camera + rear radar)

2 DETECT (MCU, every 10 ms)

SLIDE if any of:
- robot moves > 0.05 m/s for 0.3 s while tracks are stopped (RTK GNSS + IMU vs track encoders)
- slip > 30 % for 1 s while driving
- uncommanded turn > 3°/s, or heading > 5° off the fall line (the start of a spin)
- downhill acceleration > 0.5 m/s² not commanded
- roll > 12° or pitch > 33°

3 REACT (automatic, < 0.5 s)

1. Drop rear sprags + front anchors together (K10)
2. Tracks: brakes ON, never spin the tracks
3. Lower the deck if raised (CoG down)
4. Red beacon + siren; LoRa alarm to all pendants and the farmer's phone: "ROBOT SLIDING — KEEP AWAY"
5. Stay stopped. Re-arm only by a person on site

Fail-safe: power lost → magnets release → sprags drop.
False drop while climbing: arms trail, chains drag;
stop and re-arm.

4 BEST FIX: THE ROUTE

A tracked robot cannot beat wet clay at 30°,
and side-on it rolls over.
Cheapest real fix: cut a robot track on the 30°
section with a mini excavator (1–2 days):
- 1.5 m wide, ≤ 15° along the track
- level across (≤ 5°), gravel on top
- switchbacks with flat turning pads
- earth berm / kerb on the downhill edge

Level across = it can't roll even if it slides;
the anchors become a backup only.

DWG 005 REV D anti-slide 2026-10-28
PRELIMINARY — test on the real slope (BP-09 tests S1–S4)